

BASEER AHMAD PIRACHA

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PROFESSIONAL SUMMARY

Senior Embedded & Firmware Engineer with 8+ years of hands-on experience designing, developing, and deploying high-reliability embedded systems, real-time firmware, and Edge AI solutions for automotive, industrial, and telematics applications. Extensive experience in CAN bus reverse engineering and development, DBC development, and real-time vehicle data acquisition for telematics systems.

Proven expertise in Embedded C/C++ (MISRA), bare-metal programming, RTOS (FreeRTOS, Zephyr), and Embedded Linux (Yocto) on microcontrollers and platforms including ARM Cortex-M (STM32, NXP S32K), TI MSP430, ESP32, NVIDIA Jetson Orin Nano, and Kendryte CanMV K230. Strong background in low-level driver development, board bring-up, hardware-software integration, and communication protocols (CAN/CAN-FD, J1939, OBD-II, UART, SPI, I2C, MODBUS, BLE).

Skilled in secure boot, OTA updates, cryptography, and safety-critical systems compliant with ISO 26262, FIPS 140-2 and MIL-STD-461 standards. Collaborative engineer with full software lifecycle ownership — from architecture and implementation through testing, field deployment, and large-scale system integration.

KEY COMPETENCIES

Programming & Frameworks

Embedded C/C++ (MISRA), Python, FastAPI, Docker, PyTorch, TensorFlow

Embedded Systems & Platforms

RTOS (FreeRTOS), Embedded Linux (Yocto), Bare-metal, NVIDIA Jetson Orin Nano, Kendryte CanMV K230, ARM Cortex-M, STM32, NXP S32K, MSP430, ESP32

Tools & DevOps

Git, Jira, Jenkins, CI/CD, Label Studio, CVAT, Pandas, NumPy, DeepStream SDK

Communication Protocols

CAN/CAN-FD, J1939, OBD-II, LIN, MODBUS, RS485, BLE, UART, SPI, I2C

Computer Vision & Edge AI

YOLO (Training, Fine-tuning & Deployment), OpenCV, PaddleOCR, Tesseract OCR, CNNs, Video Analytics, Model Quantization, GPU Optimization, TensorFlow Lite

Automotive

CAN Bus Reverse Engineering, DBC Development, Xirgo, Galileosky, SavvyCAN, CANalyzer

Safety & Security

ISO 26262, IEC 61850, FIPS 140-2, MIL-STD-461G, Secure Boot, Cryptography (AES, SHA, HMAC)

WORK EXPERIENCE

Sr. R&D Embedded Software Engineer — AFAQY IT CO, Riyadh, Saudi Arabia

October 2025 — Present

- Designed and developed Edge AI driver management system based on Kendryte CanMV K230
- Designed, developed, and optimized real-time Edge AI computer vision solutions on NVIDIA Jetson Orin Nano using YOLO, OpenCV, and multiple OCR frameworks (PaddleOCR, Tesseract) for multi-stream video analytics including VIN detection, number plate recognition, people counting, fire & smoke detection, and parking space determination
- Built and deployed FastAPI microservices with Docker for AI model inference, integration, and backend services supporting computer vision pipelines
- Led reverse engineering of CAN bus messages for Toyota, Hyundai, MG, Chevrolet, Ford and other vehicles; developed custom DBC files for telematics integration
- Performed CAN traffic sniffing, signal decoding, and protocol analysis using CANalyzer, SavvyCAN, Vehicle Spy, and custom Python tools
- Developed embedded software and real-time data processing logic using C and Python
- Designed, validated, and integrated embedded communication systems (CAN, CAN-FD, BLE, UART) for automotive IoT applications

- Developed Lua/Galileosky scripts for harsh braking, geofencing, and advanced fleet management logic
- Integrated BLE sensors and Xirgo/Galileosky telematics units with cloud platform
- Led large-scale field deployments and system integration across Saudi Arabia
- Prepared technical documentation, decoding reports, and automated batch image processing pipelines using Pandas and NumPy

Sr. Firmware Engineer — Innovative Systems, Riyadh, Saudi Arabia

August 2023 — September 2025

- Designed and developed real-time firmware for protection relays on TI MSP430 microcontrollers compliant with international safety standards
- Implemented ANSI 49/50/51/52 protection algorithms, event logging, fault recording (oscillography), and real-time digital control logic
- Developed USB bootloader and remote firmware update application
- Integrated industrial communication protocols (MODBUS RTU/ASCII, RS485)
- Performed comprehensive testing and validation on live power systems
- Conducted low-level debugging, performance optimization, and root-cause analysis
- Worked on IEC 61850-compliant systems with strict timing and resource constraints

Sr. Embedded Software Engineer — Krämer Automotive Systems GmbH, Islamabad, Pakistan

September 2022 — August 2023

- Developed firmware for infotainment systems in Bentley and e.Go vehicles on NXP S32K and STM32 microcontrollers
- Implemented robust CAN communication stacks with timeout mechanisms and diagnostics for instrument clusters
- Designed and developed secure boot firmware and USB bootloader for OTA/USB updates
- Optimized drivers and middleware for memory, display, motor control, and HVAC systems
- Worked on Android-based automotive infotainment system integration and customization
- Performed PCB debugging, hardware validation, HIL testing, and ISO 26262 compliance activities (FMEA, safety mechanisms)
- Managed project lifecycle using Jira, Confluence, Git, and SVN

Embedded Software Lead — Crypto Research and Development Center, Islamabad, Pakistan

May 2021 — August 2022

- Led full lifecycle design and development of secure embedded systems and cryptographic modules
- Mentored junior engineers and coordinated cross-functional teams (hardware, firmware, RF, QA)
- Implemented cryptographic algorithms (AES-256, SHA-256, HMAC) and secure boot mechanisms
- Conducted reverse engineering of communication protocols using analyzers and SDR tools
- Oversaw compliance with FIPS 140-2, Common Criteria, and MIL-STD-461G standards
- Delivered technical presentations to stakeholders and managed project schedules & deliverables

Embedded Software Engineer — Crypto Research and Development Center, Islamabad, Pakistan

April 2019 — May 2021

- Designed and implemented secure embedded firmware on AVR, ESP32, STM32, and SAM platforms
- Developed secure audio communication protocols over I2S with cryptographic integration
- Integrated peripherals (SPI, UART, I2C, CAN, RTC, Flash, EEPROM) with low-power techniques
- Designed and debugged multi-layer PCBs using Altium and OrCAD
- Deployed TensorFlow Lite AI models on embedded Linux platforms (Raspberry Pi)
- Produced documentation and validation reports for FIPS 140-2 and MIL-STD compliance

EDUCATION

Master of Science in Electrical Engineering — National University of Sciences & Technology (NUST), Pakistan

September 2017 — August 2020

- GPA: 3.3/4.0
- First Class Honors
- Machine Learning and Pattern Recognition

Bachelor of Science in Electrical Engineering — COMSATS University, Pakistan

February 2013 — March 2017

- GPA: 3.41/4.0
- First Class Honors
- Final Year Project: Implementation of OFDM based Transceiver using USRP in LabVIEW

TECHNICAL SKILLS

Programming & Frameworks

Embedded C/C++ (MISRA), Python, FastAPI, FreeRTOS, Bare-metal, Yocto Linux

Microcontrollers & Platforms

ARM Cortex-M, STM32, NXP S32K, TI MSP430, AVR, ESP32, SAM, NVIDIA Jetson Orin Nano, Raspberry Pi, Kendryte CanMV K230

Communication Protocols

CAN/CAN-FD, J1939, OBD-II, LIN, MODBUS, RS485, BLE, UART, SPI, I2C, I2S, MQTT

Automotive & Telematics

CAN Reverse Engineering, DBC Development, Xirgo, Galileosky, Vehicle Spy, SavvyCAN

AI & Computer Vision

YOLO, OpenCV, PaddleOCR, Tesseract OCR, TensorFlow Lite, PyTorch, CNNs, Video Analytics, Model Quantization, GPU Optimization, Pandas, NumPy

Safety & Security

ISO 26262, IEC 61850, FIPS 140-2, MIL-STD-461G, Secure Boot, Cryptography (AES, SHA, HMAC)

Tools & DevOps

CANalyzer, Git, SVN, Jenkins, GitHub Actions, Jira, Confluence, CMake, Altium, OpenOCD, Label Studio, CVAT

Testing & Debugging

HIL Testing, Oscilloscopes, Logic Analyzers, JTAG/SWD, Protocol Analyzers

LANGUAGES

English (C1 Professional) Urdu (Native)

References available upon request